

Spectral Decomposition

Linear Algebra

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Invariant Subspaces

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eigenvectorsUpper Triangular
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Normal Operators

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Definition

Let $\mathcal{V}$ be an inner product space. Let $\mathcal{M}$ be any subspace of $\mathcal{V}$. Denote by $\mathcal{M}^\perp$ the set of all vectors in $\mathcal{V}$ orthogonal to all vectors in $\mathcal{M}$. $\mathcal{M}^\perp$ (pronounced “em perp”) is called the **orthogonal complement** of $\mathcal{M}$.

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Proposition 5.1

$\mathcal{M}^\perp$ is a subspace of $\mathcal{V}$.

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Theorem 5.2

If $\mathcal{V}$ is an inner product space, then $\mathcal{V} = \mathcal{M} \oplus \mathcal{M}^\perp$ for any subspace $\mathcal{M}$. Furthermore, the projection operators corresponding to $\mathcal{M}$ and $\mathcal{M}^\perp$ are hermitian.

Remark

Starting with a basis of a subspace $\mathcal{M}$, we can extend it to a basis $B = \{|a_i\rangle\}_{i=1}^N$ of $\mathcal{V}$ whose first m vectors span $\mathcal{M}$. The matrix representation of $\mathbf{A}$ in such a basis is given by the relation $\mathbf{A}|a_i\rangle = \alpha_{ji}|a_j\rangle$, $i = 1, 2, \dots, N$. If $i \leq m$, then $\alpha_{ji} = 0$ for $j > m$. Thus, the matrix representation of $\mathbf{A}$ in B will have the form

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{0}_{21} & \mathbf{A}_{22} \end{pmatrix},$$

where $\mathbf{A}_{11}$ is an $m \times m$ matrix, $\mathbf{A}_{12}$ an $m \times (N-m)$ matrix, $\mathbf{0}_{21}$ the $(N-m) \times m$ zero matrix, and $\mathbf{A}_{22}$ an $(N-m) \times (N-m)$ matrix. We say that $\mathbf{A}_{11}$ **represents** the operator $\mathbf{A}$ in the m -dimensional subspace $\mathcal{M}$. It may also happen that the subspace spanned by the remaining basis vectors in B is also an invariant subspace of $\mathbf{A}$. Then $\mathbf{A}_{12}$ will be zero, and $\mathbf{A}$ will take a **block diagonal** form:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix}.$$

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Definition

*If a matrix representing an operator can be brought into a block diagonal form by a suitable choice of basis, it is called **reducible**; otherwise it is called **irreducible**. A reducible matrix A is denoted in two different ways:*

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} \iff A = A_1 \oplus A_2.$$

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Definition

A subspace $\mathcal{M}$ is an **invariant subspace** of the operator $\mathbf{A}$ if $\mathbf{A}(\mathcal{M}) \subset \mathcal{M}$. We say that $\mathcal{M}$ **reduces** $\mathbf{A}$ if both $\mathcal{M}$ and $\mathcal{M}^\perp$ are invariant subspaces of $\mathbf{A}$.

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Lemma 5.3

A subspace $\mathcal{M}$ of an inner product space $\mathcal{V}$ is invariant under the linear operator $\mathbf{A}$ if and only if $\mathcal{M}^\perp$ is invariant under $\mathbf{A}^\dagger$.

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Theorem 5.4

A subspace of $\mathcal{V}$ reduces $\mathbf{A}$ if and only if it is invariant under both $\mathbf{A}$ and $\mathbf{A}^\dagger$.

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Lemma 5.5

Let $\mathcal{M}$ be a subspace of $\mathcal{V}$ and $\mathbf{P}$ the hermitian projection operator onto $\mathcal{M}$. Then $\mathcal{M}$ is invariant under the linear operator $\mathbf{A}$ if and only if $\mathbf{AP} = \mathbf{PAP}$.

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Theorem 5.6

Let $\mathcal{M}$ be a subspace of $\mathcal{V}$, $\mathbf{P}$ the hermitian projection operator of $\mathcal{V}$ onto $\mathcal{M}$, and $\mathbf{A}$ the linear operator on $\mathcal{V}$. Then $\mathcal{M}$ reduces $\mathbf{A}$ if and only if $\mathbf{A}$ and $\mathbf{P}$ commute.

Definition

Let $\mathbf{A} \in \text{End}(\mathcal{V})$ be a linear operator, and $|a\rangle$ a nonzero vector such that

$$\mathbf{A}|a\rangle = \lambda|a\rangle, \quad \lambda \in \mathbb{C}.$$

We then say that $|a\rangle$ is an **eigenvector** of $\mathbf{A}$ with **eigenvalue** $\lambda^\dagger$.

[†]Sometimes, the abuse of notation $|\lambda\rangle$ is used to denote both the facts that $|\lambda\rangle$ is an eigenvector of an operator (which is usually understood from the context) and that its eigenvalue is λ .

Proposition 5.7

Add the zero vector to the set of all eigenvectors of $\mathbf{A}$ belonging to the same eigenvalue λ , and denote the span of the resulting set by $\mathcal{M}_\lambda$. Then $\mathcal{M}_\lambda$ is a subspace of $\mathcal{V}$, and every (nonzero) vector in $\mathcal{M}_\lambda$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ .

Definition

The subspace $\mathcal{M}_\lambda$ is referred to as the **eigenspace** of $\mathbf{A}$ corresponding to the eigenvalue λ . Its dimension is called the **geometric multiplicity** of λ . An eigenvalue is called **simple** if its geometric multiplicity is 1. The set of eigenvalues of $\mathbf{A}$ is called the **spectrum** of $\mathbf{A}$.

Theorem 5.8

If $\{\lambda_i\}_{i=1}^r$ are distinct eigenvalues of an operator $\mathbf{A}$ and $\mathcal{M}_i$ is the eigenspace corresponding to λ_i , then

$$\sum_{i=1}^r \mathcal{M}_i = \bigoplus_{i=1}^r \mathcal{M}_i.$$

Remark

Rewriting the definition of eigenvector and eigenvalue and using an abuse of notation, we have

$$(\mathbf{A} - \lambda \mathbf{1})|\lambda\rangle = |0\rangle.$$

That is, $|\lambda\rangle$ is an eigenvector of $\mathbf{A}$ if and only if $|\lambda\rangle \in \ker(\mathbf{A} - \lambda \mathbf{1})$. We know from Theorem 1.15 and Theorem 1.17 (Dimension Theorem) that if $\mathbf{A} - \lambda \mathbf{1}$ is invertible (bijective) then its kernel consists only on the zero vector, which is not acceptable as a solution. Hence, if we are to obtain nontrivial solution, $\mathbf{A} - \lambda \mathbf{1}$ must have no inverse, and, as we saw in the determinants section, this is true if and only if

$$\det(\mathbf{A} - \lambda \mathbf{1}) = 0.$$

Definition

The determinant $\det(\mathbf{A} - \lambda \mathbf{1})$ is a polynomial in λ , called the **characteristic polynomial** of $\mathbf{A}$. The roots of this polynomial are called **characteristic roots** and are simply the eigenvalues of $\mathbf{A}$.

Proposition 5.9

Every operator on a finite-dimensional linear space over $\mathbb{C}$ has at least one eigenvalue and therefore at least one eigenvector.

Definition

Let $\lambda_{j=1}^p$ be the distinct roots of the characteristic polynomial of $\mathbf{A}$, and let λ_j occur m_j times. Then m_j is called the **algebraic multiplicity** of λ_j , and

$$\det(\mathbf{A} - \lambda \mathbf{1}) = \prod_{j=1}^p (\lambda_j - \lambda)^{m_j}.$$

For $\lambda = 0$ we find the determinant of $\mathbf{A}$ to be

$$\det \mathbf{A} = \prod_{j=1}^p \lambda_j^{m_j}.$$

Example 5.10

Find the eigenvalues of the matrix A defined as

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and determine the algebraic and geometric multiplicity of each eigenvalue.

Proposition 5.11

An operator is invertible if and only if none of its eigenvalues is zero.

Example 5.12

Find the eigenvalues of a projection operator $\mathbf{P}$.

Definition

A linear operator $\mathbf{A}$ on a linear space $\mathcal{V}$ is said to be **diagonalisable** if there is a basis for $\mathcal{V}$ all of whose vectors are eigenvectors of $\mathbf{A}$.

Theorem 5.13

Let $\mathbf{A}$ be a diagonalisable operator on a linear space $\mathcal{V}$ with distinct eigenvalues $\{\lambda_j\}_{j=1}^r$. Then there are idempotents $\mathbf{P}_j$ on $\mathcal{V}$ such that

$$\text{i) } \mathbf{1} = \sum_{j=1}^r \mathbf{P}_j,$$

$$\text{ii) } \mathbf{P}_i \mathbf{P}_j = 0 \text{ for } i \neq j, \text{ and}$$

$$\text{iii) } \mathbf{A} = \sum_{j=1}^r \lambda_j \mathbf{P}_j.$$

Theorem 5.14

An operator $\mathbf{T}$ is invertible if and only if its upper-triangular matrix representation has no zero on its main diagonal. The entries on the main diagonal are simply the eigenvalues of $\mathbf{T}$.

Theorem 5.15

Let $\mathcal{V}$ be a complex linear space of dimension N and $\mathbf{T} \in \text{End}(\mathcal{V})$. Then there exists a basis of $\mathcal{V}$ in which $\mathbf{T}$ is represented by an upper triangular matrix.

Definition

A **normal operator** is an operator on an inner product space that commutes with its adjoint.

Proposition 5.16

Each of the following types of operators are normal:

- *hermitian;*
- *anti-hermitian;*
- *unitary;*
- *real and orthogonal;*
- *real and symmetric;*
- *real and antisymmetric.*

Proposition 5.17

The operator $\mathbf{A} \in \text{End}(\mathcal{V})$ satisfies

$$\|\mathbf{A}x\| = \|\mathbf{A}^\dagger x\| \quad \text{for all } |x\rangle \in \mathcal{V}$$

if and only if $\mathbf{A}$ is normal.

Theorem 5.18

Let $\mathbf{A}$ be a normal operator on $\mathcal{V}$ and $\mathcal{U}$ a subspace of $\mathcal{V}$ invariant under $\mathbf{A}$. Then $\mathcal{U}$ is invariant under $\mathbf{A}^\dagger$. Therefore by Theorem 5.4 any invariant subspace of a normal operator reduces it.

Theorem 5.19

Let $\mathbf{A}$ be a normal operator on $\mathcal{V}$. Then $|\lambda\rangle$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ if and only if $|x\rangle$ is an eigenvector of $\mathbf{A}^\dagger$ with eigenvalue λ^ .*

Corollary 5.20

The eigenvalues of a hermitian operator are real. A unitary operator has unimodular eigenvalues.

Example 5.21

Find the eigenvalues and (normalised) eigenvectors of the matrix

$$H = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Theorem 5.22

An eigenspace of a normal operator reduces that operator. Moreover, eigenspaces of a normal operator are mutually orthogonal.

Theorem 5.23 (Complex Spectral Decomposition)

Let $\mathbf{A}$ be a normal operator on a finite-dimensional complex inner product space $\mathcal{V}$. Let $\lambda_1, \lambda_2, \dots, \lambda_r$ be its distinct eigenvalues. Then

$$\mathcal{V} = \bigoplus_{j=1}^r \mathcal{M}_j,$$

and the nonzero (hermitian) projection operators $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_r$, where $\mathbf{P}_j$ projects onto $\mathcal{M}_j$, form a complete set of orthogonal idempotents which also satisfy

$$\mathbf{A} = \sum_{j=1}^r \lambda_j \mathbf{P}_j.$$

Proposition 5.24

If $\mathbf{A} \in \text{End}(\mathcal{V})$ is normal, then $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$. Therefore, a normal operator on a complex inner product space is diagonalisable.

Corollary 5.25

A hermitian operator is positive if and only if all its eigenvalues are positive.

Theorem 5.26

An operator on a finite-dimensional complex inner product space is normal if and only if it is diagonalisable.

Theorem 5.27

A hermitian matrix can always be brought to diagonal form by means of a unitary transformation matrix.

Definition

*Two operators are said to be **simultaneously diagonalisable** if they can be written in terms of the same set of projection operators, as in Theorem 5.20 (Complex Spectral Decomposition)*

Lemma 5.28

An operator $\mathbf{T}$ commutes with a normal operator $\mathbf{A}$ if and only if $\mathbf{T}$ commutes with all the projection operators of $\mathbf{A}$.

Theorem 5.29

Two normal operators $\mathbf{A}$ and $\mathbf{B}$ are simultaneously diagonalisable if and only if $[\mathbf{A}, \mathbf{B}] = 0$.

Example 5.30

Let us find the spectral decomposition of the Pauli spin matrix

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Remark

Functions of operators were discussed in Section 1 (Linear Spaces). We begin by noting that if $\mathbf{T} = \lambda_i \mathbf{P}_i$, then $f(\mathbf{T}) = f(\lambda_i) \mathbf{P}_i$ for any f analytic.

Example 5.31

Investigate the spectral decomposition of the following unitary (actually orthogonal) matrix:

$$U = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Theorem 5.32

A unitary operator $\mathbf{U}$ in a finite-dimensional complex inner product space can be written as $\mathbf{U} = e^{i\mathbf{H}}$ where $\mathbf{H}$ is hermitian. Furthermore, a unitary matrix can be brought to diagonal form by a unitary transformation matrix.

Definition

The **positive square root** of a positive (thus hermitian, thus normal) operator $\mathbf{A} = \lambda_i \mathbf{P}_i$ is $\sqrt{\mathbf{A}} = \sqrt{\lambda_i} \mathbf{P}_i$.

Example 5.33

Evaluate $\sqrt{\mathbf{A}}$ where

$$\mathbf{A} = \begin{pmatrix} 5 & 3i \\ -3i & 5 \end{pmatrix}.$$

Theorem 5.34

Let $\mathcal{V}$ be a finite-dimensional linear space and $\mathbf{T} \in \text{End}(\mathcal{V})$ a normal operator. Then

$$f(\mathbf{T}) = \sum_{j=1}^r f(\lambda_j) \mathbf{P}_j = \sum_{j=1}^r f(\lambda_j) \prod_{k \neq j}^r \frac{\mathbf{T} - \lambda_k \mathbf{1}}{\lambda_j - \lambda_k},$$

i.e., every function of $\mathbf{T}$ is a polynomial.

Theorem 5.35

An operator on a real linear space has invariant subspace of dimension 1 or 2.

Theorem 5.36

Every operator on an odd-dimensional real linear space has a real eigenvalue and an associated eigenvector.

Lemma 5.37

Let $\mathbf{T}$ be a self-adjoint (hermitian) operator on a linear space $\mathcal{V}$. Then

$$\mathbf{H} \equiv \mathbf{T}^2 + \alpha \mathbf{T} + \beta \mathbf{1}, \quad \alpha, \beta \in \mathbb{R}, \alpha^2 < 4\beta,$$

is invertible.

Theorem 5.38

A self-adjoint (symmetric) real operator has a real eigenvalue.

Theorem 5.39

Let $\mathcal{V}$ be a real inner product space and $\mathbf{T}$ a self-adjoint operator on $\mathcal{V}$. Then there exists an orthonormal basis in $\mathcal{V}$ with respect to which $\mathbf{T}$ is represented by a diagonal matrix.

Theorem 5.40

Let $\mathcal{V}$ be a real linear space and $\mathbf{T}$ a normal operator on $\mathcal{V}$. Let $\{\lambda_i\}_{i=1}^r$ be a complete set of the distinct eigenvalues of $\mathbf{T}$ and $\{(\alpha_j, \beta_j)\}_{j=1}^s$ all the distinct pairs labeling the second degree polynomials of equation. Let $\mathcal{M}_{\lambda_i} = \ker p_{\lambda_i}(\mathbf{T})$ and $\mathcal{M}_{\alpha_j, \beta_j} = \ker p_{\alpha_j, \beta_j}(\mathbf{T})$. Then

$$\mathcal{V} = \left(\bigoplus_{i=1}^r \mathcal{M} \right) \oplus \left(\bigoplus_{j=1}^s \mathcal{M}_{\alpha_j, \beta_j} \right),$$

where $\lambda_i, \alpha_j, \beta_j \in \mathbb{R}$ and $\alpha_j^2 < 4\beta_j$.

Theorem 5.41 (Real Spectral Decomposition)

Let $\mathcal{V}$ be a real linear space and $\mathbf{T}$ a normal operator on $\mathcal{V}$. Let $|x_i\rangle$ and $|y_j\rangle$ satisfy equations 1, and 2, respectively. Then

$$\mathcal{V} = \left(\bigoplus_{i=1}^M \text{span}\{|x_i\rangle\} \right) \oplus \left(\bigoplus_{j=1}^K \text{span}\{|y_j\rangle, \mathbf{T}|y_j\rangle\} \right)$$

with $\dim \mathcal{V} = 2K + M$.

Theorem 5.42

A real orthogonal operator on a real inner product space $\mathcal{V}$ cannot, in general, be completely diagonalised. The closest it can get to a diagonal form is

$$O_{diag} = \text{diag}(\underbrace{1, 1, \dots, 1}_{N_+}, \underbrace{-1, -1, \dots, -1}_{N_-}, R_2(\theta_1), R_2(\theta_2), \dots, R_2(\theta_m)),$$

where $N_+ + N_- + 2m = \dim \mathcal{V}$ and $R_2(\theta_j)$ is

$$R_2(\theta_j) = \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \cos \theta_j & \sin \theta_j \\ \sin \theta_j & -\cos \theta_j \end{pmatrix}.$$

Furthermore, the matrix that transforms an orthogonal matrix into the form above is itself an orthogonal matrix.

Theorem 5.43 (Euler's Theorem)

The general motion of a rigid body consists of the translation of one point of that body and a rotation about a single axis through that point.

Theorem 5.44

The determinant of a positive definite matrix M is given by

$$\det M = \frac{\pi^n}{(\int d^n x \exp(-x^t M x))^2}$$

Theorem 5.45 (Polar Decomposition)

An operator $\mathbf{T}$ on a (real or complex) finite-dimensional inner product space can be written as $\mathbf{T} = \mathbf{U}\mathbf{R}$ where $\mathbf{R}$ is a positive operator and $\mathbf{U}$ is an isometry.

Example 5.46

Find the polar decomposition of

$$A = \begin{pmatrix} -2i & \sqrt{7} \\ 0 & 3 \end{pmatrix}.$$

Example 5.47

Decompose the following real matrix into its polar form.

$$A = \begin{pmatrix} 2 & 0 \\ 3 & -1 \end{pmatrix}.$$